

SALES PERFORMANCE ANALYSIS REPORT

Alfido Tech Internship

Submitted By

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1. Objective

The main objective of this project was to analyze sales performance using a retail Superstore dataset. I wanted to understand how sales varied across different product categories, customer segments, regions, and states. By cleaning the dataset, exploring the data, and creating visualizations, I was able to identify important sales patterns and gain insights that could help support better business decisions.

2. Dataset Description

For this project, I used the **Superstore Sales Dataset** available on Kaggle. The dataset contains nearly **9,800 sales records** collected from a retail business and includes information about customer orders, products, sales, shipping details, and locations.

The dataset contains the following important attributes:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Customer Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales

The dataset provides detailed information about customer orders and sales, making it suitable for understanding sales performance across different products, customer groups, and regions.

3. Data Cleaning and Preprocessing

Before starting the analysis, I carefully examined the dataset to understand its structure and quality.

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked the number of rows and columns.
- Verified column names.
- Examined data types.
- Checked for missing values.
- Checked for duplicate records.
- Converted the Order Date and Ship Date columns into datetime format.
- Prepared the dataset for further analysis.

The dataset was already well organised, so only a few preprocessing steps were needed. Most of the work involved checking the data, converting the date columns into the correct format, and preparing the dataset for analysis.

4. Feature Engineering

To make the analysis more meaningful, a few additional features were created from the existing data.

The **Order Date** column was used to extract the **Year** and **Month**, making it easier to study sales trends over different time periods.

These additional columns simplified the analysis and made the visualizations easier to understand.

5. Exploratory Data Analysis

After cleaning the dataset, I explored it using summary statistics and different visualizations.

Statistical measures such as mean, median, mode, minimum, maximum, and standard deviation were calculated for the numerical columns.

The exploratory analysis mainly focused on:

- Product categories
- Customer segments
- Sales performance
- Regional sales
- Monthly sales trends
- Product-wise sales

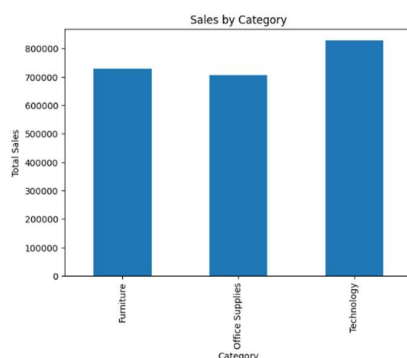
Exploring the dataset using summary statistics gave me a better understanding of the sales data before creating visualizations. This made it easier to identify meaningful trends during the analysis.

6. Data Visualization

Several visualizations were created to better understand sales performance.

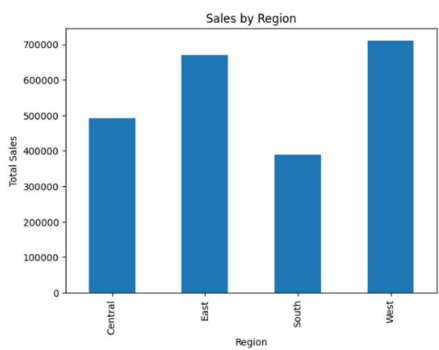
Sales by Category

A bar chart was created to compare total sales across different product categories. This chart helped me understand which categories contributed more to the overall sales revenue.



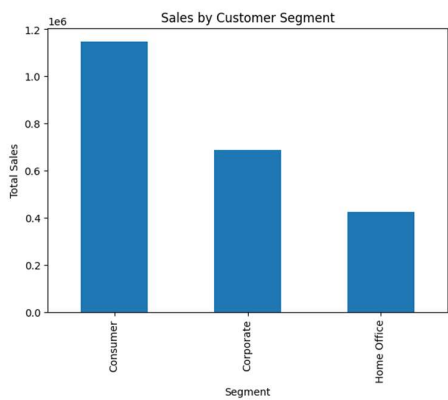
Sales by Region

This visualization compares total sales across different regions. It provides a clear overview of regional sales performance and highlights areas with stronger business activity.



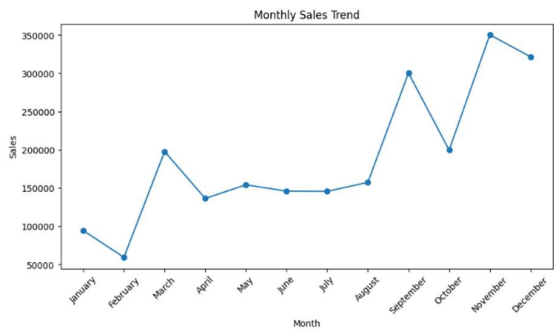
Sales by Customer Segment

A bar chart was created to compare sales generated by different customer segments. It helped identify which customer groups contributed more to total sales.



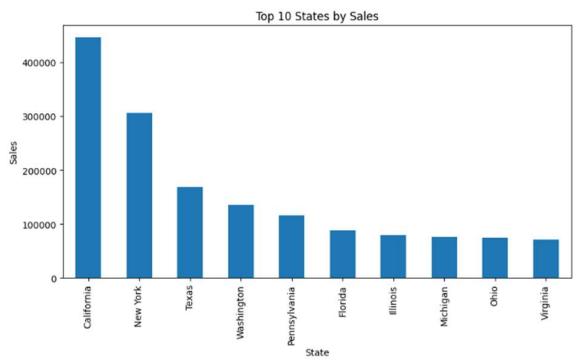
Monthly Sales Trend

This chart gives a simple overview of how sales varied across different months. It helps in understanding monthly purchasing patterns throughout the year.



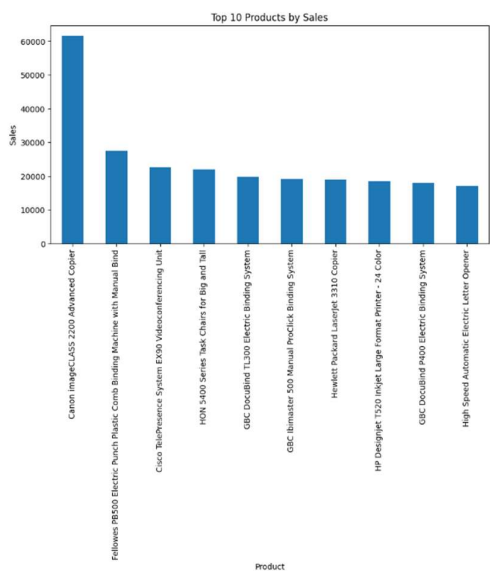
Top 10 States by Sales

This visualization highlights the ten states with the highest sales. It provides useful information about the locations contributing the most to business revenue.



Top 10 Products by Sales

A bar chart was created to compare the products generating the highest sales. This helps identify products that contribute significantly to overall business performance.



7. Sales Performance Analysis

After analysing the dataset, I noticed that sales differed across product categories, customer segments, and regions. The analysis showed that some product categories and regions generated higher sales than others. It also highlighted a few products that contributed significantly to the overall revenue, making them important for business growth.

Along with the visual analysis, I calculated key performance indicators such as **Total Revenue**, **Average Order Value**, and **Total Orders** to better understand the overall sales performance. These KPIs provided a simple summary of the business performance and supported the insights obtained from the visualizations.

The monthly sales trend also showed that sales changed throughout the year, indicating variations in customer purchasing behaviour over time.

8. Key Findings

After analysing the dataset and the generated visualizations, I observed several important patterns.

- The dataset contains nearly 9,800 sales records suitable for business analysis.
- Sales varied across different product categories.
- Regional sales performance differed noticeably.
- Customer segments contributed differently to total sales.
- Monthly sales showed changes throughout the year.
- A few products generated significantly higher sales than others.
- Key performance indicators such as Total Revenue, Average Order Value, and Total Orders provided a quick overview of the overall sales performance.

9. Recommendations

Based on the analysis, I believe businesses can improve their sales performance by focusing on their strongest products, customer segments, and regions.

The following recommendations may help improve business performance:

- Increase inventory for products with consistently higher sales.
- Develop region-specific marketing strategies to strengthen sales performance.
- Introduce customer loyalty programs for valuable customer segments.
- Plan promotional campaigns during periods of lower sales.
- Monitor product and regional performance regularly to support better business decisions.

10. Conclusion

Working on this project gave me practical experience in analysing retail sales data using Python.

During this project, I learned how to clean a real-world sales dataset, explore the data using summary statistics, create meaningful visualizations, and interpret the results. Overall, this project improved my understanding of sales analysis and gave me more confidence in working with business datasets using Python.
